# Difficulty Analysis of Posiform-Planted QUBO Benchmarks via Energy and Hamming Metrics

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***Abstract*—Benchmarking QUBO solvers requires instances with a known optimum that stay hard. Posiform planting builds such instances with a unique optimum. A recent variant adds random spin-glass terms and scales the planted term by a coefficient $\alpha$ to tune hardness. Prior work characterizes this hardness only empirically, via the success rate, which shows that difficulty varies with the parameters but not why, and cannot compare instances at $0\%$ success. We instead measure how close failing runs come to the optimum. Over failures only, on a fixed hardest-instance set, we record the energy gap and Hamming distance (HD) to the planted answer. These resolve the flat $0\%$ into a continuous difficulty curve with a clear structure: $\Delta E = \Delta R{+}\alpha\Delta P$, a local-minimum floor plus a growing planted penalty. Under simulated annealing (SA) on a D-Wave Pegasus graph ($n$=5612) it is non-monotone in $\alpha$: $\Delta E$ peaks then declines while HD falls throughout, with the peak where the rising penalty is overtaken by failures drifting toward the optimum. A D-Wave Advantage run reproduces the shape, suggesting an instance property, not SA alone. Reporting both quantities grades planted QUBO benchmarks even when no run succeeds.**

***Index Terms*—QUBO benchmarking, posiform planting, quantum annealing, simulated annealing, Hamming distance, energy gap, $\alpha$-difficulty.**

## I. Introduction

Benchmarking quantum annealers and classical heuristics on QUBO problems needs instances whose optima are known but hard to find. Posiform planting [1] produces QUBOs with a unique ground state, but they are easy for simulated annealing (SA) [2]. Pelofske et al. [3] made them harder. They add random discrete-coefficient spin-glass QUBOs $R_i$ to a posiform-planted QUBO $P$ with unique minimum $x^\star$, scaled by a coefficient $\alpha$:

$$Q_\alpha(x) \;=\; \sum_i R_i(x) \;+\; \alpha\, P(x). \tag{1}$$

That study tuned hardness mainly through the random-QUBO size and the annealing time. It fixed $\alpha \in \{0.01, 0.1\}$ and reported the ground-state success rate. But this rate is $0\%$ across the whole small-$\alpha$ regime. It cannot rank instances, solvers, or $\alpha$ values. Pelofske et al. conjecture that the hardness comes from local minima whose energy is very close to the true ground state.

We make that conjecture quantitative. $\Delta E$ measures local-minimum depth. But over all samples it mixes depth with the success rate. We therefore condition on failure and use the 100 hardest instances, held fixed as $\alpha$ varies. Measured this way, failure-conditioned $\Delta E$ and Hamming distance (HD) reveal a $\Delta R{+}\alpha\Delta P$ structure with a non-monotone peak. Neither quantity is new. Our contribution is the protocol. Failure-conditioning on a fixed hardest set turns the uninformative $0\%$ rate into a rankable curve. A supporting quantum-annealer run is consistent with this structure coming from the instances, not SA alone.

## II. Methodology

**Instances.** We generate 500 posiform-planted instances on the D-Wave Pegasus P16 graph [4] ($n$=5612 qubits). Each coefficient $R_{ij}$ is $-1$ or $+1$. We split the graph into blocks of at most 16 qubits by recursive bisection and solve each block $R_i$ exactly. The planted target $x^\star$ is the concatenation of the block ground states, so it is the global minimum of $\sum_i R_i$. $P$ is the posiform of a planted 2-SAT whose only solution is $x^\star$. We rank the 500 instances by their total SA failing-reads across the sweep and fix the hardest 100 as a constant set. This set is chosen from SA alone, before any QPU run, so the cross-solver agreement is not circular.

**Metrics.** For each sample $x$ we record HD$(x, x^\star)$ [5] and the energy gap $\Delta E = E(x) - E(x^\star) \geq 0$ under the same $Q_\alpha$. Because every $R_{ij}$ is $\pm 1$, each $R$ energy is an integer. We therefore report $\Delta E$ in units of the smallest non-degenerate $R$ gap ($\Delta_R$=1). This unit is the same across instances, as is HD$/n$. A read counts as a success only when it equals $x^\star$ (a bit match), so the $0\%$ success rate and the per-read failure rate both track $x^\star$ exactly. This differs from matching the ground-state energy, which at $\alpha$=0 is degenerate (Sec. III-A). At large $\alpha$, correct samples ($\Delta E{\approx}0$, HD=0) drive both quantities to 0. We therefore condition all statistics below on failure ($x{\neq}x^\star$). Means are over failing reads. The instance-level standard error stays below $0.06\,\Delta_R$, so we omit error bars.

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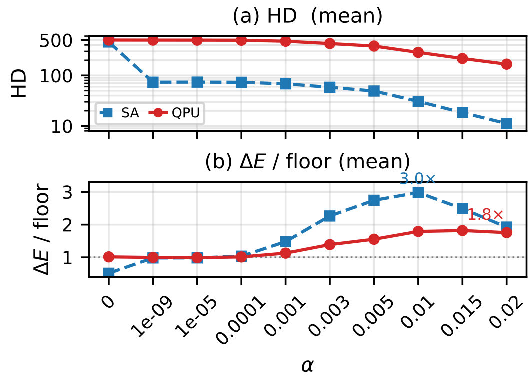


Fig. 1. Failure-conditioned difficulty over the 100 hardest instances. At each $\alpha$, at least 91% of reads fail. Blue: SA. Red: a D-Wave Advantage annealer. (a) HD on a log axis falls monotonically but never reaches 0. (b) $\Delta E$ rescaled by each solver's small-$\alpha$ floor (SA $\approx 1.1\,\Delta_R$, QPU $\approx 24$), on one axis. Both start at 1. SA rises to a peak of 3.0× near $\alpha{=}10^{-2}$, then declines. The annealer shows the same shape but more weakly (1.8×).

**Protocol.** We sweep $\alpha$ over 0, $10^{-9}$, $10^{-5}$, $10^{-4}$, $10^{-3}$, $3{\times}10^{-3}$, $5{\times}10^{-3}$, $10^{-2}$, $1.5{\times}10^{-2}$, and $2{\times}10^{-2}$. Beyond $2{\times}10^{-2}$ the instances are increasingly solved, so no fixed failing set remains. The main sweep uses the `neal` SA package [6] at 1000 sweeps and 100 reads per (instance, $\alpha$); we also run a separate 150-instance subset (not the hardest-100) at 1000 and 5000 sweeps to compare budgets. For a cross-solver check we replay the same 100 instances on a D-Wave Advantage annealer (`Advantage_system6.4`). The run is hardware-native, with auto-scaling, two-gauge averaging, and 100 reads at 20 $\mu$s. We recompute $\Delta E$ from $Q_\alpha$.

## III. Results

### A. Floor, peak, and decline of $\Delta E$

At $\alpha{=}0$ the random part $R$ is degenerate. Here 56.1% of samples reach an $R$ ground state ($\Delta E{=}0$), yet 0% equal $x^\star$, and HD is maximal (459, or 8.2% of $n$). So $\Delta E$ alone would read as solved, but HD shows no sample is near $x^\star$. For $\alpha{>}0$ (Fig. 1b), in the small-$\alpha$ region ($\alpha \in [10^{-9}, 10^{-4}]$), $\alpha\Delta P$ is negligible and $\Delta E$ stays flat at the depth of a local-minimum trap in $R$. We call this flat value the floor (mean 1.10, median 1). This floor of $\approx 1\,\Delta_R$ is the quantitative content of the local-minimum conjecture in [3]: in the small-$\alpha$ limit, failing samples sit about one $R$-gap above $x^\star$. HD is also flat ($\approx$ 73). Past $\alpha{=}10^{-3}$ the planted term grows. $\Delta E$ rises to a peak of 3.35 at $\alpha{=}10^{-2}$, which is 3.0× the floor, then declines to 2.16. The floor, peak, and decline are separated by far more than the instance-level SE ($< 0.06\,\Delta_R$). HD meanwhile falls monotonically (73 → 11). So failing samples move closer to $x^\star$ in bits even as their energy gap grows. The peak comes from a competition. For a fixed failing bitstring, $\Delta E{=}\Delta R{+}\alpha\Delta P$ is linear in $\alpha$ ($\Delta R, \Delta P$ fixed per bitstring). But the failing set is not fixed: as $\alpha$ rises, failing samples drift toward $x^\star$ (HD falls), where $\Delta P$ is smaller. The observed mean is an envelope over this moving population: it climbs while $\alpha\Delta P$ dominates, then falls as the drift into smaller-$\Delta P$ regions wins. The peak is not a survivorship or selection artifact. The 100 instances fail at least 91% of reads at every $\alpha$ (0% ground-state success, as [3] report). The same shape appears over all 500 instances (floor $\approx$ 1.1, peak $\approx$ 3.0), and the hardest set only sharpens it.

### B. Quantum-annealer consistency check

As supporting evidence, we ran the same 100 instances once on a D-Wave Advantage annealer (Fig. 1, red). It shows the same pattern. There is 0% success, a non-monotone $\Delta E$ peak near $\alpha{=}10^{-2}$, and a monotonic HD decrease (501 → 167). Its baselines sit an order of magnitude above SA's ($\Delta E \approx 24$, HD$\approx$ 500), and its peak is weaker in relative terms (1.8× its floor versus SA's 3.0×). One run cannot establish generality. But it is consistent with the structure being a property of the instances, not of SA. The integrated control error (ICE) is the annealer's control-precision limit: couplers carry a roughly fixed error of order 0.02–0.03. At the peak the planted couplers map to Ising $J \approx 0.03$, comparable to ICE, so the device only partly resolves the signal, consistent with the weaker peak above. A smaller scale would push them below ICE and should flatten it, which we leave to future work.

## IV. Discussion and Conclusion

Failure-conditioned $\Delta E$ measures difficulty (local-minimum depth) and HD measures proximity to $x^\star$. Reporting both separates how deep a failure is from how close it is. The all-sample mean (the success rate) hides the peak. It falls at larger $\alpha$ only because more reads succeed, not because failures are shallower. $\Delta E$ also separates solvers that the success rate cannot: on the 150-instance subset at $\alpha{=}10^{-3}$, both budgets give 0% success, yet $\Delta E$ is 1.68 versus 1.00. We therefore recommend reporting failure-conditioned $\Delta E$ (floor and peak) with HD over a fixed hardest set whenever an $\alpha$-like parameter is swept. This gives a failure-conditioned diagnostic of the benchmark instances rather than a single difficulty score. Absolute values still depend on the solver and schedule.

**Limitations.** The QPU evidence is a single run (two gauges, one 20 $\mu$s anneal). The classical side is a single SA family. The instances use one coefficient type ($\pm1$) and one topology (Pegasus).

**Future work:** broader classical solvers, more annealer gauges and anneal times, other coefficient types and topologies, and joint HD–$\Delta E$ analysis. Code and instances: https://github.com/YIDEUNKIM/qubo_dataset.

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